

Optimization Techniques

Lab n°1 - Line search methods

Exercise 1 The aim of this exercise is to compare the speed of three line search algorithms, that is, the **dichotomy**, the **Golden Section search method** and the **Fibonacci search method**.

To write the programs, will use the simple function $f(x) = 8e^{1-x} + 7\log(x)$; afterwards, we will run the programs on other test functions.

1. Define the cost function

in Matlab `f = @(x) 8*exp(1-x)+7*log(x)`

in Python:

```
import numpy as np
```

```
f = lambda x: 8 * np.exp(1 - x) + 7 * np.log(x)
```

and plot it on the interval [1,2]

use the command `fplot(f, [1 2])` in Matlab

in Python you can import Matplotlib).

Is it unimodal?

2. Write a program to implement the dichotomy method that locates the minimizer of f over [1,2] to within an uncertainty of $\epsilon = 0.23$.

How many iteration are needed? How many evaluation of the function f ?

Use the Matlab commands `tic ... toc` to measure the duration of the program.

In Python, you can define

```
t0 = time.time() % at the beginning of the algorithm
```

```
% do stuff
```

```
elapsed = time.time() - t0 % at the end of the algorithm
```

3. Repeat item 2. using the Golden Section Method and the Fibonacci method.
Compare the elapsed times.

4. Repeat items 2. and 3. for the function

```
f = @(x) x.^4 - 3*x.^2 + x + 1 + 0*sum(rand(1, 1e7))
```

```
f = lambda x: x**4 - 3*x**2 + x + 1 + 0 * np.sum(np.random.rand(10**7))
```

over the interval $[-1,3]$.

What do you remark?

Exercise 2 Write a program that implements Newton-Raphson's algorithm to find the minimum of the function $f(x) = x^2 - 4x + 2$ in the interval $[-1,3]$.

1. Try the starting points $x_0 = 3, 6, 8, -15$. In how many steps does the algorithm converge ?
2. Run the same program with the function $f(x) = \frac{1}{2}x^2 - \sin(x)$, taking $x_0 = 2, 3, 0$. Does the algorithm converge?

Exercise 3 We finally consider the function $f(x) = (\arctan(x))^2$. The requested precision is 10^{-5} . Compare (elapsed time of computation, number of iterations) Newton-Raphson's algorithm with dichotomy (or golden section, or both), in the following cases:

1. Starting interval for dichotomy $[0, 0.5]$, starting point for Newton-Raphson's $x_0 = 0.25$.
2. Starting interval for dichotomy $[0, 3]$, starting point for Newton-Raphson's $x_0 = 1.5$.